COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE031	**SECTION:**	CPE21S1
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ACTIVITY NO. 13 **Designing a Purpose-Driven Data Dashboard**

1. PROCEDURE OUTPUT

What procedures did you perform?

Task 1

EXPLANATION

PANGILINAN:
->This dashboard shows how students are performing across subjects in one simple view. It compares average scores, score spread, and attendance so teachers can quickly see which subjects need more support. It also checks whether better attendance goes with higher scores. The goal is to help guide better teaching decisions.

PASION:
-> The dashboard visually presents the student populous data through graphs and charts. It shows the Student, Subject, Grade and Attendance record. The purpose of this graph is to map out the capabilities of the students, their tendency to be present and the possible correlation of the two. This can be used for teachers and faculties to improve their curriculum, making of schedules, and setting up co-curricular programs.

TORRES:
-> The purpose of this Student Performance Dashboard created in the process of this activity's procedure is to serve as an informative tool for tracking the students' performance in classes such as Math, Science, English, Programming, and Data Analytics. By visualizing and analyzing the data in the graphs and charts, it becomes possible to observe patterns and changes in students' performance and the level of engagement in classes. For Example, by looking at the Subject Average Score and Score Distribution charts, one can compare the performance in different areas of study. Moreover, with the help of the Attendance vs. Performance chart, it is possible to determine whether there is a connection between attendance rates and students' scores. In conclusion, the data dashboard can be utilized to make informed decisions about the observed process and get a better understanding of students' performance in the context of the given classes.

Task 2

CODE

```
# Aggregated data
avg_scores = df.groupby("Subject", as_index=False)["Score"].mean()
avg_attendance = df.groupby("Subject", as_index=False)["Attendance (%)"].mean()

# Dashboard layout
fig = make_subplots(rows=2, cols=2,
                    subplot_titles=("Average Score per Subject", "Attendance vs. Score",
                                    "Score Distribution", "Average Attendance per Subject"),
                    specs=[[{"type": "bar"}, {"type": "scatter"}],
                           [{"type": "box"}, {"type": "bar"}]])

fig.add_trace(go.Bar(x=avg_scores["Subject"], y=avg_scores["Score"], name="Average Score"), row=1, col=1)
fig.add_hline(y=THRESHOLD, line_dash="dash", line_color="red", row=1, col=1)

# Scatter split into normal vs. underperforming
fig.add_trace(go.Scatter(x=ok["Attendance (%)"], y=ok["Score"], mode="markers",
                         marker=dict(color="steelblue"), name="Score ≥ 75",
                         text=ok["Student"]), row=1, col=2)
fig.add_trace(go.Scatter(x=low["Attendance (%)"], y=low["Score"], mode="markers",
                         marker=dict(color="red", symbol="x", size=9),
                         name="Underperforming (< 75)", text=low["Student"]), row=1, col=2)
fig.add_hline(y=THRESHOLD, line_dash="dash", line_color="red", row=1, col=2)

fig.add_trace(go.Box(x=df["Subject"], y=df["Score"], name="Score Distribution"), row=2, col=1)
fig.add_trace(go.Scatter(x=low["Subject"], y=low["Score"], mode="markers",
                         marker=dict(color="red", size=7), name="Underperforming",
                         showlegend=False, text=low["Student"]), row=2, col=1)
fig.add_hline(y=THRESHOLD, line_dash="dash", line_color="red", row=2, col=1)

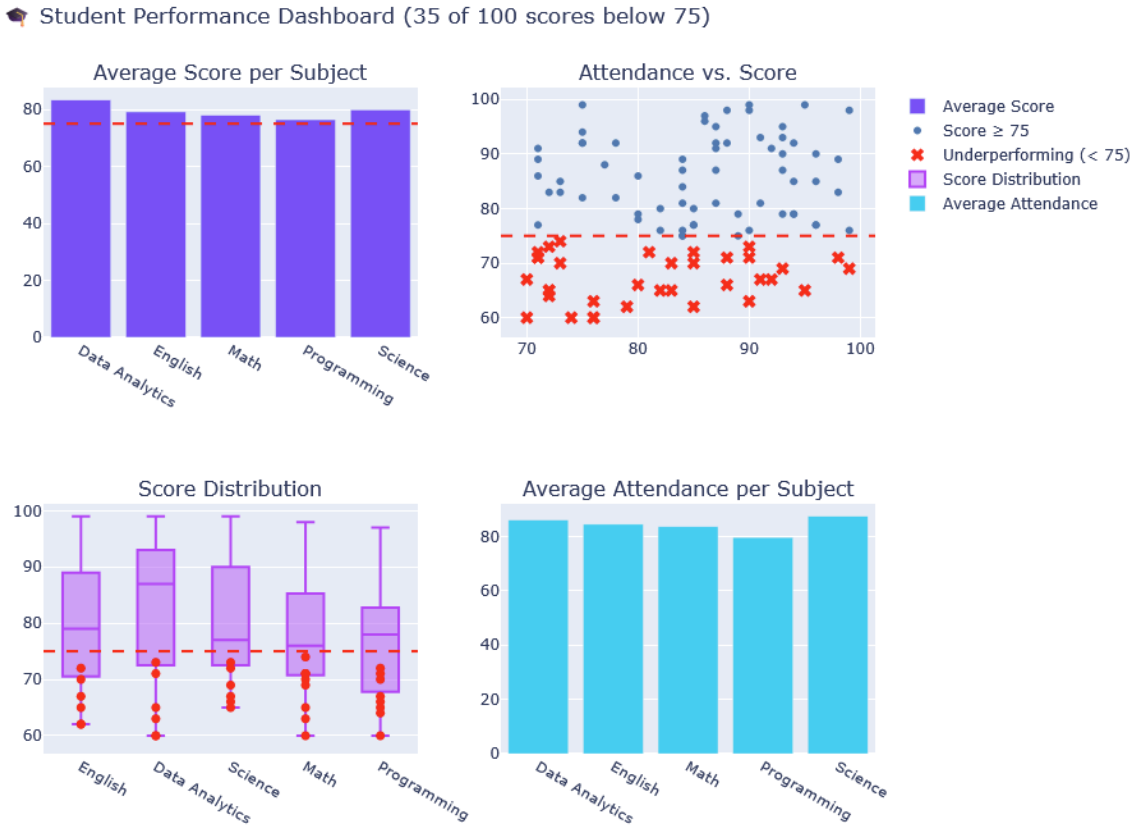
fig.add_trace(go.Bar(x=avg_attendance["Subject"], y=avg_attendance["Attendance (%)"], name="Average Attendance"), row=2, col=2)

fig.update_layout(height=700, width=950,
                  title_text=f"📊 Student Performance Dashboard ({len(low)} of {len(df)} scores below {THRESHOLD})")
fig.show()
```

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ANALYSIS

PANGILINAN:
Because we presented the data for Task 2 from the perspective of a teacher, 35 out of 100 scores are failing, and scores below the 75 passing threshold are highlighted in red. Every subject has an average above 75, with programming having the lowest score at roughly 76 and data analytics having the highest at roughly 83. Programming requires the most assistance because it also has the lowest attendance. Nearly all attendance levels have failing scores, demonstrating that good attendance does not ensure a passing grade. All subjects have some failing students, but programming has the lowest overall scores, followed by data analytics and English.

PASION:
In task 2 we've chosen to present the data in what a teacher would want to see, the student's performances. Here we explicitly highlighted the failing grade, which in this scenario is 74 and below. For the Average score per Subject bar chart, we can conclude that the programming subject has students that the majority are approximately close to a failing grade. In the scatter plot, we can see that even though the attendance rate

(x-axis) is high, we still get failing grades from students. And lastly for the Box-and-Whisker plot, we can see the outliers of failing grades from all the subjects in red.

TORRES:
For Task 2, we presented the data from a teacher's perspective, focusing on student performance and clearly marking failing grades (74 and below). The bar chart of average scores shows that Programming students tend to hover near the failing line. The scatter plot reveals that high attendance doesn't always translate to passing grades, as some students with strong attendance still fail. Finally, the box-and-whisker plot flags failing-grade outliers in red for every subject.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

CODE

```python
import numpy as np
import pandas as pd
import plotly.graph_objects as go
from plotly.subplots import make_subplots

# synthetic data: hourly energy (kWh) for 7 days, 4 campus areas
np.random.seed(42)
loads = {"Laboratory": (12, 40), "Office": (6, 28), "Classroom": (4, 32), "Library": (5, 23)}  # (off-hours, daytime)

df_energy = pd.DataFrame(
    [(day, hour, area, (load[1] if 8 <= hour < 18 else load[0]) * np.random.uniform(0.9, 1.1))
     for day in range(7) for hour in range(24) for area, load in loads.items()],
    columns=["Day", "Hour", "Area", "Energy_kWh"])
df_energy["Period"] = np.where(df_energy["Hour"].between(8, 17), "Daytime (8AM-6PM)", "Off-hours")

# for bar chart
area_energy = df_energy.groupby("Area", as_index=False)["Energy_kWh"].sum()

# for pie chart
period_energy = df_energy.groupby("Period", as_index=False)["Energy_kWh"].sum()

# for line chart
hourly_energy = df_energy.groupby(["Hour", "Area"], as_index=False)["Energy_kWh"].mean()

# Create dashboard layout
fig = make_subplots(rows=2, cols=2,
                    subplot_titles=("Total Weekly Energy per Area (kWh)", "Energy Share by Time of Day", "Average Hourly Energy per Area (kWh)"),
                    specs=[[{"type": "bar"}, {"type": "pie"}],
                           [{"colspan": 2}, None]])

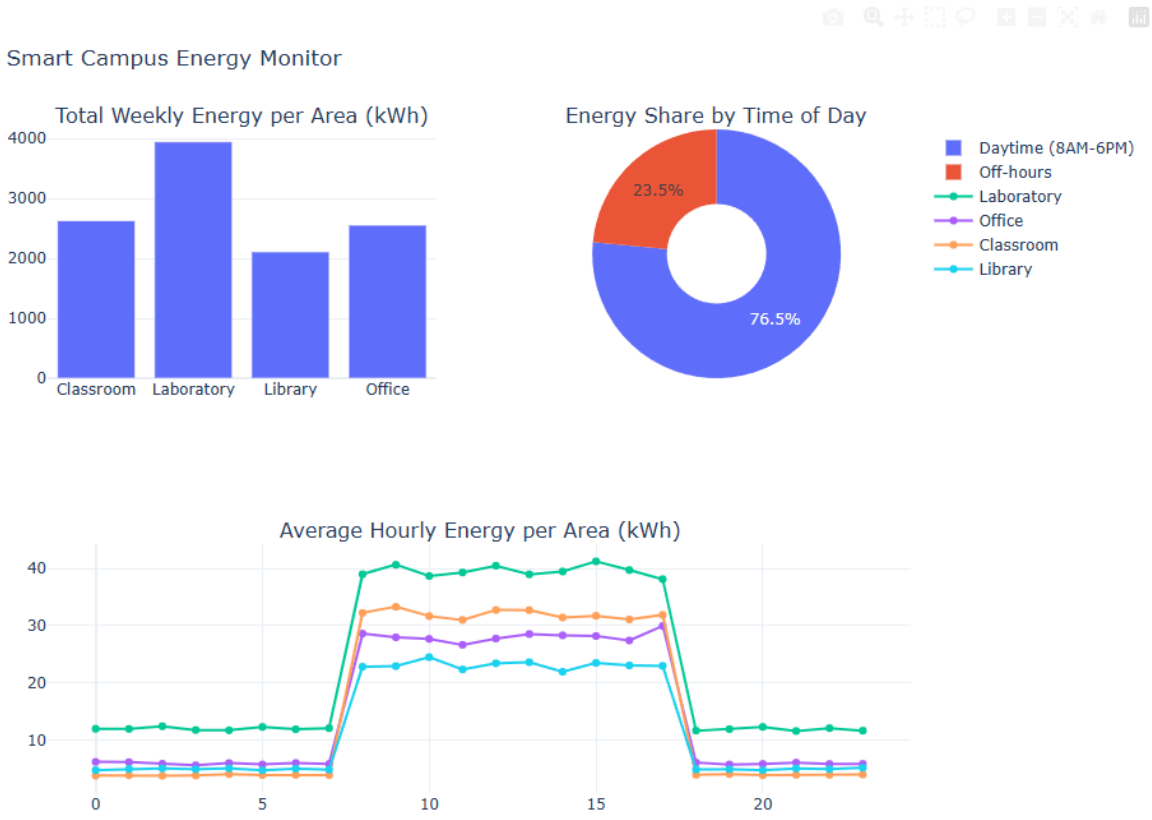
# Add charts
fig.add_trace(go.Bar(x=area_energy["Area"], y=area_energy["Energy_kWh"], name="Total Energy", showlegend=False), row=1, col=1)
fig.add_trace(go.Pie(labels=period_energy["Period"], values=period_energy["Energy_kWh"], hole=0.4, name="Time of Day"), row=1, col=2)
for area in loads:
    d = hourly_energy[hourly_energy["Area"] == area]
    fig.add_trace(go.Scatter(x=d["Hour"], y=d["Energy_kWh"], mode="lines+markers", name=area), row=2, col=1)

fig.update_layout(height=700, width=950, title_text="Smart Campus Energy Monitor", template="plotly_white")
fig.show()
```

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ANALYSIS

PANGILINAN:

-> We did this so every element helps answer where and when energy is being wasted. The layout moves from overview to detail. Each chart type matches its question, line for change over time, bar for comparing totals, pie for share of the whole, and heatmap for patterns by day and hour. Each area keeps the same color in every chart so the legend only needs to be learned once, and a red scale highlights high usage and the anomaly. Interactivity is limited to hover tooltips, legend filtering, and a zoom slider, which give the user drill-down without extra controls to learn. Units, labels, and a clean white background keep the data accurate and easy to read.

PASION:

-> Here, we used a bar chart to present the average power usage per type of area, a pie chart to show the ratio of the energy share by time of day, and lastly a line graph to show the average hourly energy use per area. We chose these designs because we believe these factors hold significant impact in determining and seeing where we need maintenance, power-saving techniques, and etc.

TORRES:
-> This dashboard implements a 2×2 grid to place the two summary visualizations (the bar plot with the weekly aggregates and the donut chart that compares day versus night shares) on the top and provide the hourly line chart with the bottom row to accommodate its 24 points across four lines, which requires more space. The specs parameter is necessary to let Plotly know that it should place a pie chart with the cartesian axes plots. Each chart type was chosen according to the type of insights they were supposed to illustrate, which are bars for comparisons between groups, a donut chart for a two-group proportion analysis, and lines for depicting temporal trends. The initial code uses default colors for simplicity and applies a plotly_white style to make the background visible, while my version attempts to use colors associated with each area to make the entire dashboard easier to read since one would not have to look at the legend to understand what line or bar corresponds to which group. No interactivity was added beyond default hover effects and the ability to zoom and toggle the visibility of legend items by clicking on them. The key limitation of the dashboard is that it is static and always uses the 950×700 resolution, so it would not be easy to adjust it for different screen sizes.

CONCLUSION

PANGILINAN:
-> In this activity I learned that a good dashboard starts from its purpose and its user, not from the charts. Knowing who the dashboard is for shapes every choice, from the layout to the chart types to how much interactivity it needs. Each chart should answer one question the user actually has, so the chart type must fit that question, colors should stay consistent, and interactivity should be added only where it helps. The main limitation is that the data was simulated, so the results reflect assumptions rather than real measurements.

PASION:
-> In this Hands-On Activity, we've learned how to design a dashboard, with a purpose to solve a problem or to develop efficiency. The tasks done are aligned with allowing the viewer to understand and scope out factors that may help improve solutions. For the supplementary activity, we've simulated the data for a "mapped out" dashboard of the energy usage in the campus. This can help the electrical engineers to determine the power consumption of each building, room, and etc. In conclusion, the way you make, view, and interpret the graph can be impactful in solving and creating solutions.

TORRES:
-> In this Hands-On Activity, I've learned on how to use a Purpose-Driven Data Dashboard which are user-friendly which has different features like zooming in, zooming out, download as PNG file, and other features that are useful on understanding the dataset. In Task 1, I've learned that the data dashboard can be utilized to make informed decisions about the observed process and get a better understanding of students' performance in the context of the given classes. In the Task 2, I've learned that a dashboard should be tailored to its end user, as opposed to simply having charts. When we redesigned the dashboard for a teacher, we've added a threshold of 75 with a red dashed line and "x" marks to indicate underperformance. I noticed that the scores in Programming tended to be on the brink of failing and that high attendance did not necessarily correlate with high scores, which means that attendance cannot be the sole factor affecting performance. In the case of the Supplementary Activity, we've assigned different types of graphs to different questions which are bars for totals, a donut chart for the distribution of day and off-hour classes, and line graphs for hourly trends. From this experience, I have learned that a dashboard should be designed with an explicit focus on its intended user and purpose, and that the choice of graphs should facilitate decision-making rather than

simply display values.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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